# Product Requirements Document (PRD)

InfoBridge: Smart Access to Public Services

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## 1 Abstract

InfoBridge is a Streamlit-based Retrieval-Augmented Generation (RAG) chatbot for Indian government services. It retrieves information from official PDF documents, builds context with source citations, and generates grounded answers through a Groq-backed LLM client. The interface includes English/Hindi UI support, a collapsible sidebar, theme persistence, and service filtering.

## 2 Introduction

Government service information is often distributed across long PDF manuals and circulars. InfoBridge addresses this by allowing users to ask service-related questions in a conversational interface and receive answers that are grounded in the indexed documents.

## 3 Objectives

- Provide a searchable chatbot for government service information.
- Use a RAG pipeline to keep answers grounded in official PDFs.
- Show source citations for every assistant response.
- Support English and Hindi UI labels.
- Provide a clean Streamlit experience with theme switching and a collapsible sidebar.

## 4 Scope

**Included services:**

- Passport Services
- Voter ID / EPIC
- Driving Licence & RC
- Income Tax
- Ayushman Bharat

**Out of scope:**

- Real-time government APIs
- Authentication
- Transaction processing

# 5 System Overview

The current system follows the pipeline below:

1. PDFs are extracted from the service folders under `data/`.
2. Documents are chunked into overlapping text segments.
3. Chunks are embedded with `sentence-transformers/all-MiniLM-L6-v2`.
4. Embeddings and metadata are stored in FAISS.
5. A user query is embedded and searched against the vector store.
6. Retrieved chunks are formatted into context and citations.
7. The Groq client generates the final answer.
8. The UI renders the answer and source list in Streamlit.

# 6 Functional Requirements

- Accept queries through a Streamlit chat interface.
- Retrieve relevant document chunks using semantic search.
- Generate grounded answers with conversation memory.
- Display source citations for each answer.
- Filter retrieval by service category.
- Support English and Hindi UI labels.
- Persist theme selection and sidebar open/closed state within the session.

# 7 Non-Functional Requirements

- Fast and responsive chat interaction.
- Reliable answer grounding through retrieved context.
- Clear readability in both light and dark modes.
- Maintainable modular Python codebase.

## 8 Technology Stack

- Frontend: Streamlit
- Backend: Python
- Embeddings: Sentence Transformers MiniLM
- Vector Store: FAISS
- LLM: Groq chat completions
- PDF Parsing: PyPDF2
- Chunking: LangChain text splitters

## 9 Current UI Behavior

- The sidebar can collapse into a left strip.
- The app remembers the selected theme across reruns.
- The language selector is English/Hindi only.
- The service filter uses translated labels.
- Assistant responses render source citations below the answer.

## 10 Implementation Notes

- API key is read from the `API`$_{K}EY environmentvariable.$
- `The default model is llama3-8b-8192 with fallback model support.`
- `The translation table is stored in frontend/translations_en_hi.csv.`
- `Session state tracks messages, memory, language, theme, and sidebar state.`

## 11 Verification

- Build the FAISS index using `python scripts/build_index.py`.
- Run the app using `streamlit run app.py`.
- Validate that answers include sources and that the sidebar/theme state persists.

## 12 Conclusion

InfoBridge is now implemented as a complete modular RAG chatbot with a current Streamlit UI, Groq-based generation, persistent theme handling, a collapsible sidebar, and bilingual interface labels.